ปีที่ 1 ภาคการศึกษาที่ 1			
รหัสวิชา	**รายวิชา**		**หน่วยกิต**
2110104*	การทำโปรแกรมคอมพิวเตอร์	3	(3-0-6)
2110222*	บทนำสู่วิศวกรรมคอมพิวเตอร์และเทคโนโลยีดิจิทัล	3	(3-0-6)
2110328*	โครงสร้างข้อมูลและขั้นตอนวิธี	4	(4-0-8)
2110204*	โครงสร้างดิสครีต	3	(3-0-6)
2110252*	ตรรกศาสตร์ของดิจิทัลคอมพิวเตอร์	3	(3-0-6)
	ปีที่ 2 ภาคการศึกษาที่ 1		
2110205	สถิติสำหรับวิศวกรรมคอมพิวเตอร์	3	(3-0-6)
2110201	คณิตศาสตร์วิศวกรรมคอมพิวเตอร์	3	(3-0-6)
2110203	คณิตศาสตร์วิศวกรรมคอมพิวเตอร์ 2	3	(3-0-6)
5500203	ภาษาอังกฤษที่จำเป็นสำหรับวิศวกร	3	(3-0-6)
2110403	วิทยาศาสตร์ข้อมูลและวิศวกรรมข้อมูลเบื้องต้น	3	(3-0-6)
2110356	ระบบฝังตัว	3	(3-0-6)

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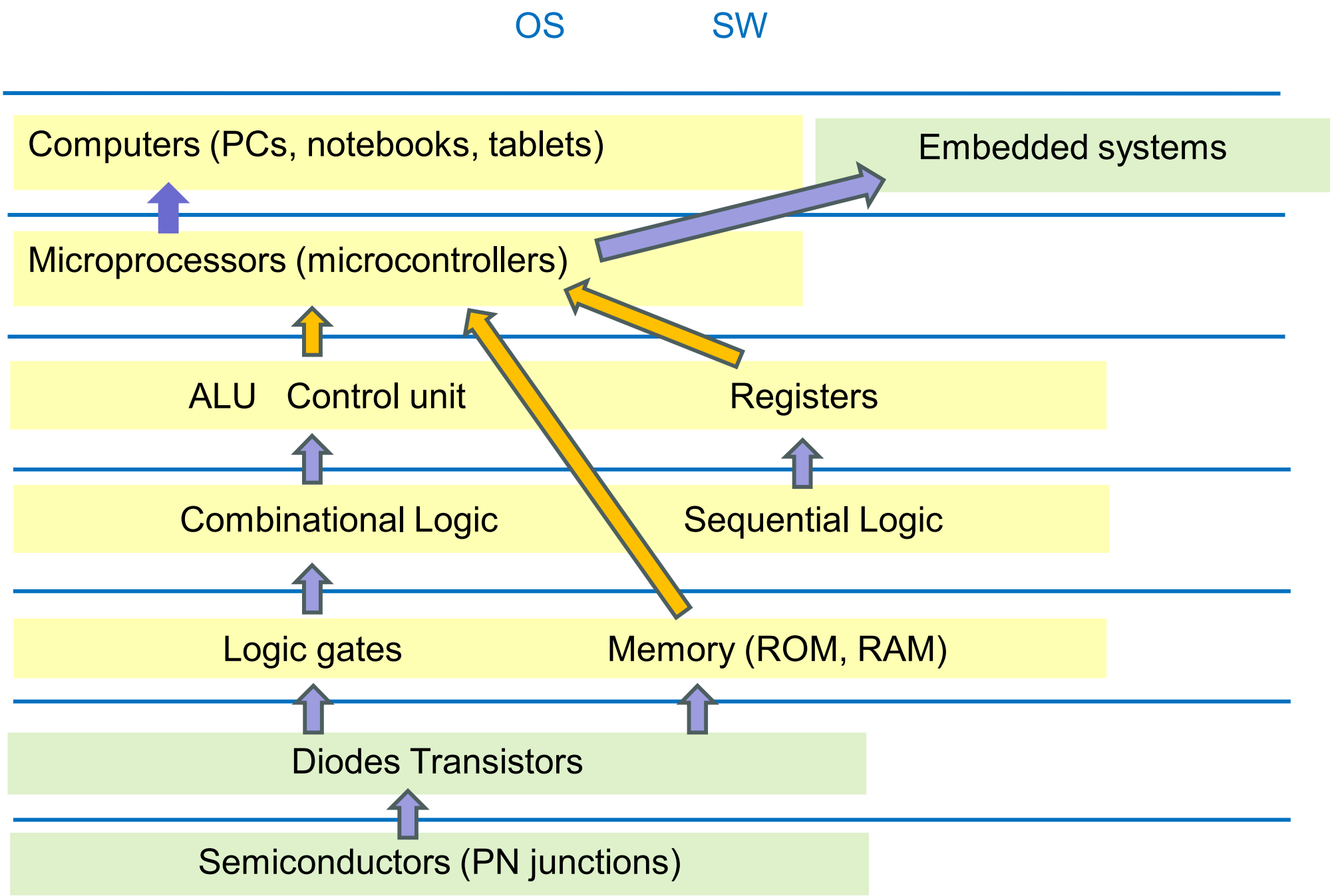
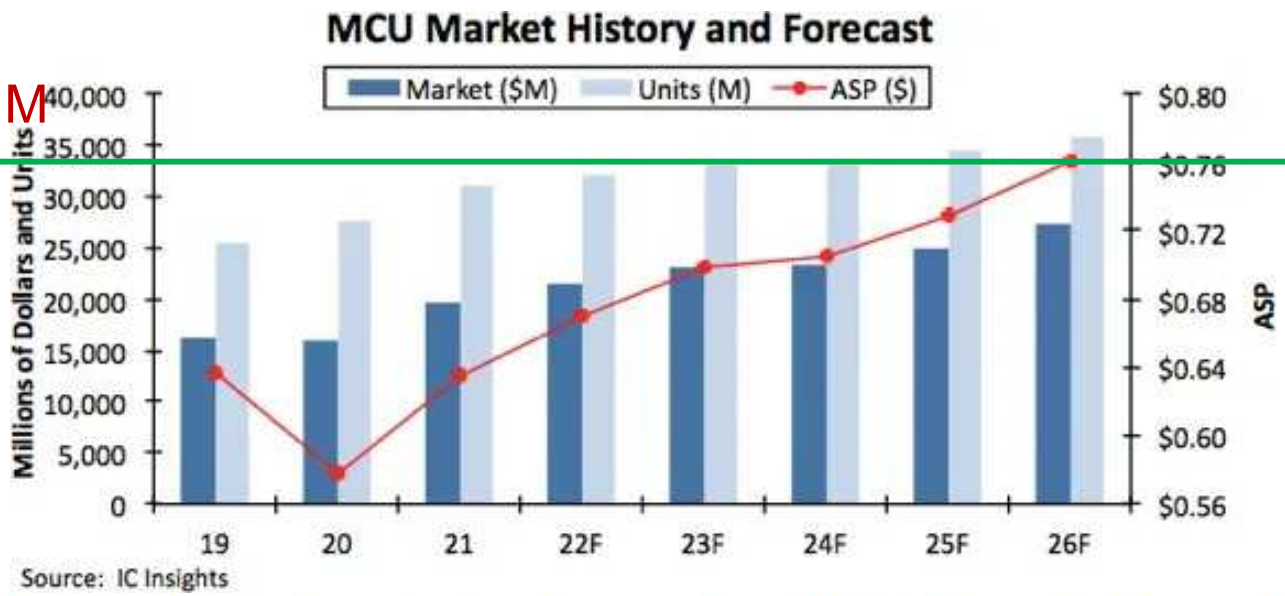
HW classes


OS
SW
Computers (PCs, notebooks, tablets)
Embedded systems
Microprocessors (microcontrollers)
ALU
Control unit
Registers
Combinational Logic
Sequential Logic
Logic gates
Memory (ROM, RAM)
Diodes Transistors
Semiconductors (PN junctions)

Interesting information

34000 M

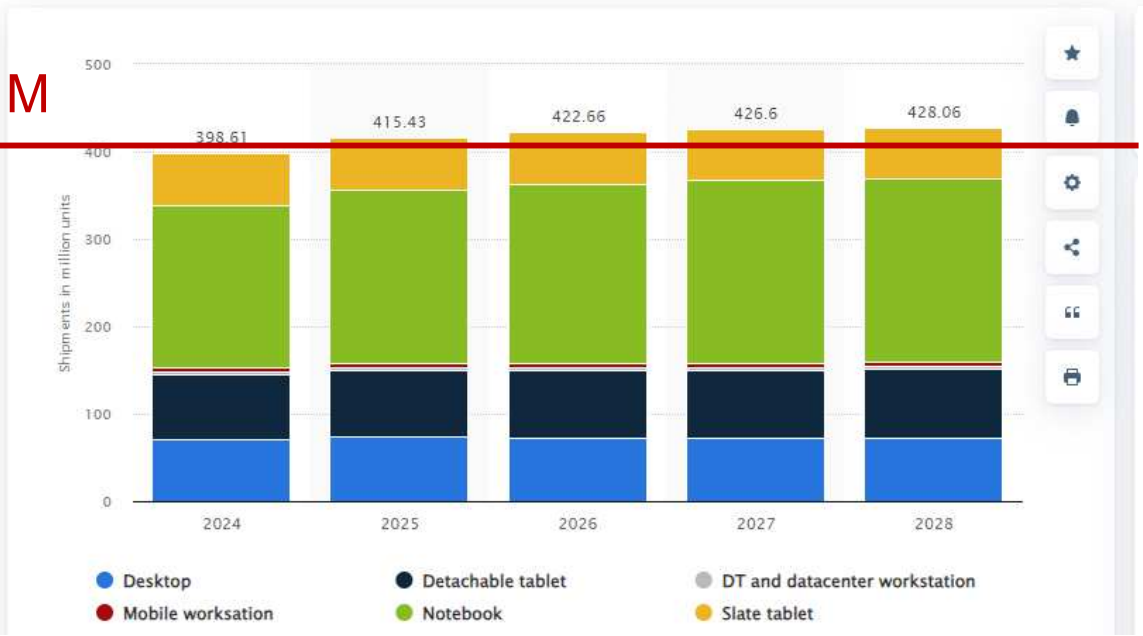


Source: https://www.eetasia.com/mcus-get-a-lift-from-automotive-after-2021-rebound/

Personal computing devices shipments worldwide from 2024 to 2028

(in million units)

400 M



Source: https://www.statista.com/statistics/272595/global-shipments-forecast-for-tablets-laptops-and-desktop-pcs/

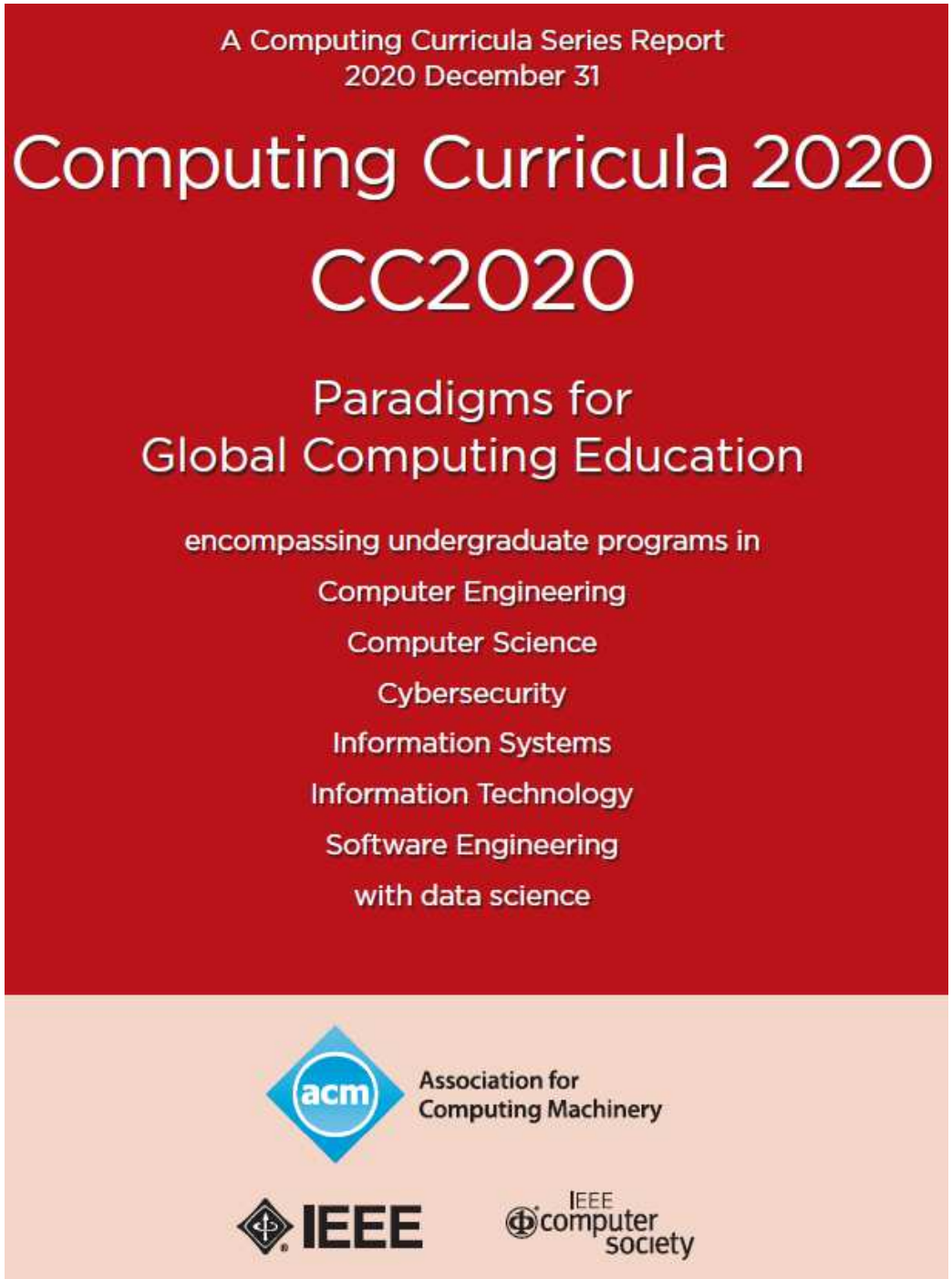
From https://www.electronicshub.org/embedded-system-real-time-applications/

Examples of Embedded Systems

Many Different Products Depend on Embedded Systems

From: Embedded Systems Trends and Technologies, DICK SLANSKY, 2018 (www.arcweb.com)

Domain Knowledge in computing



Domain Knowledge in computing

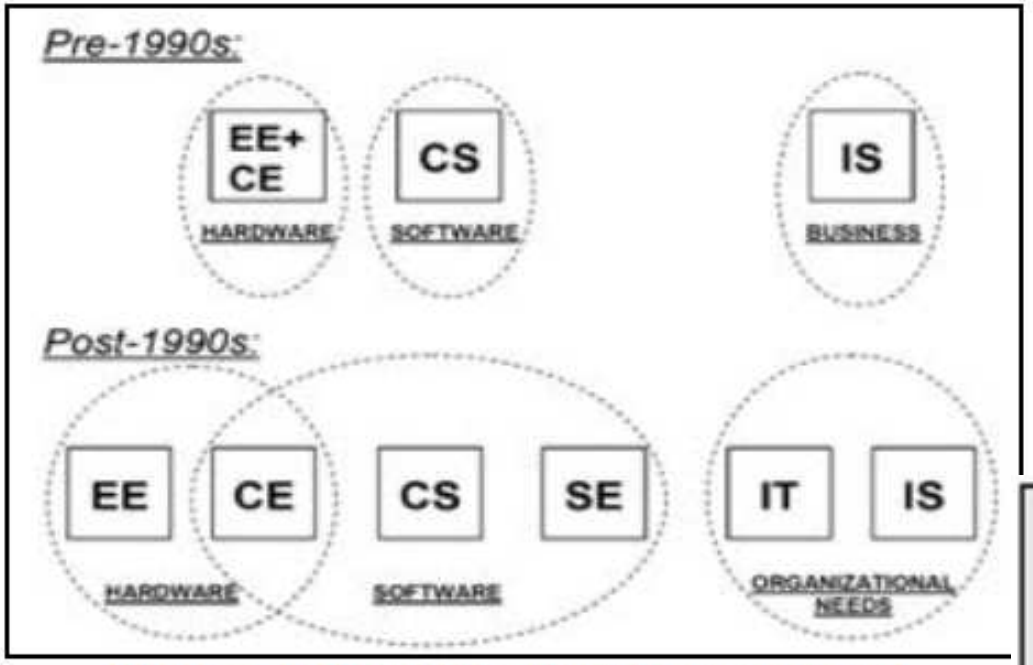
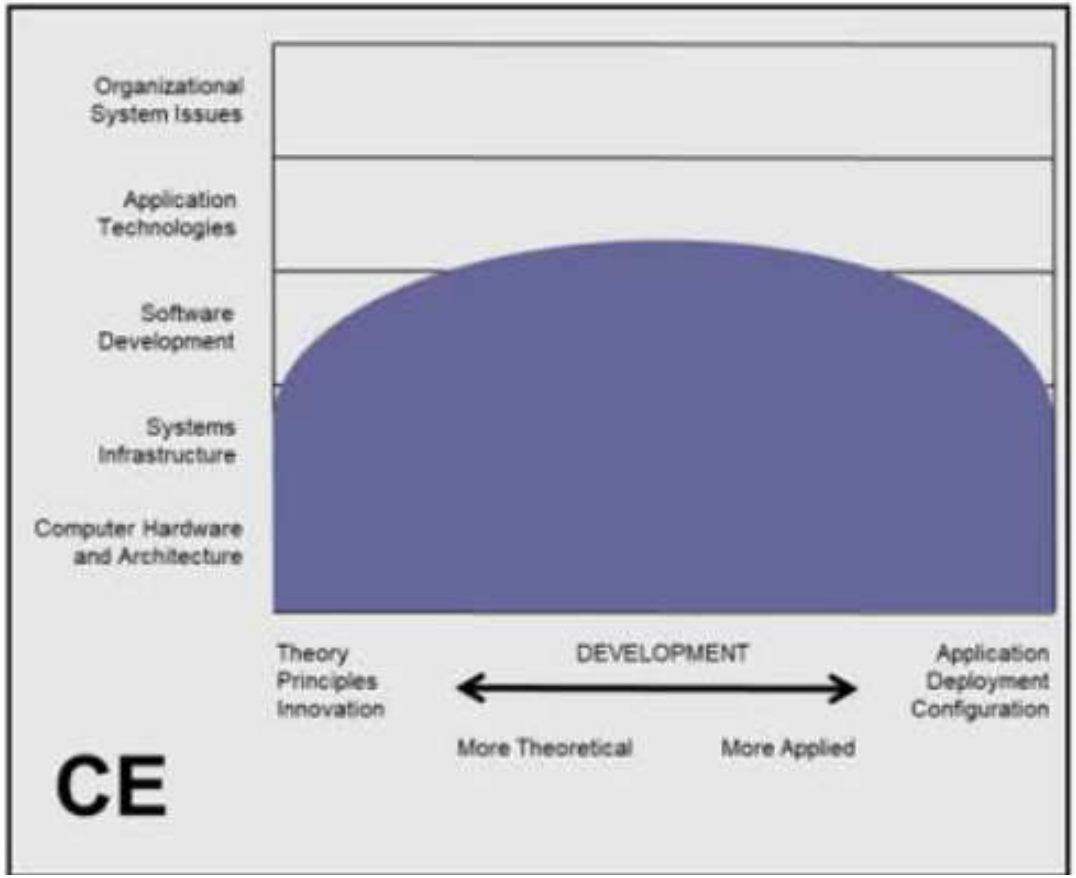


Figure 2.1 Computing disciplines compared, from CC2005



Domain Knowledge in computing

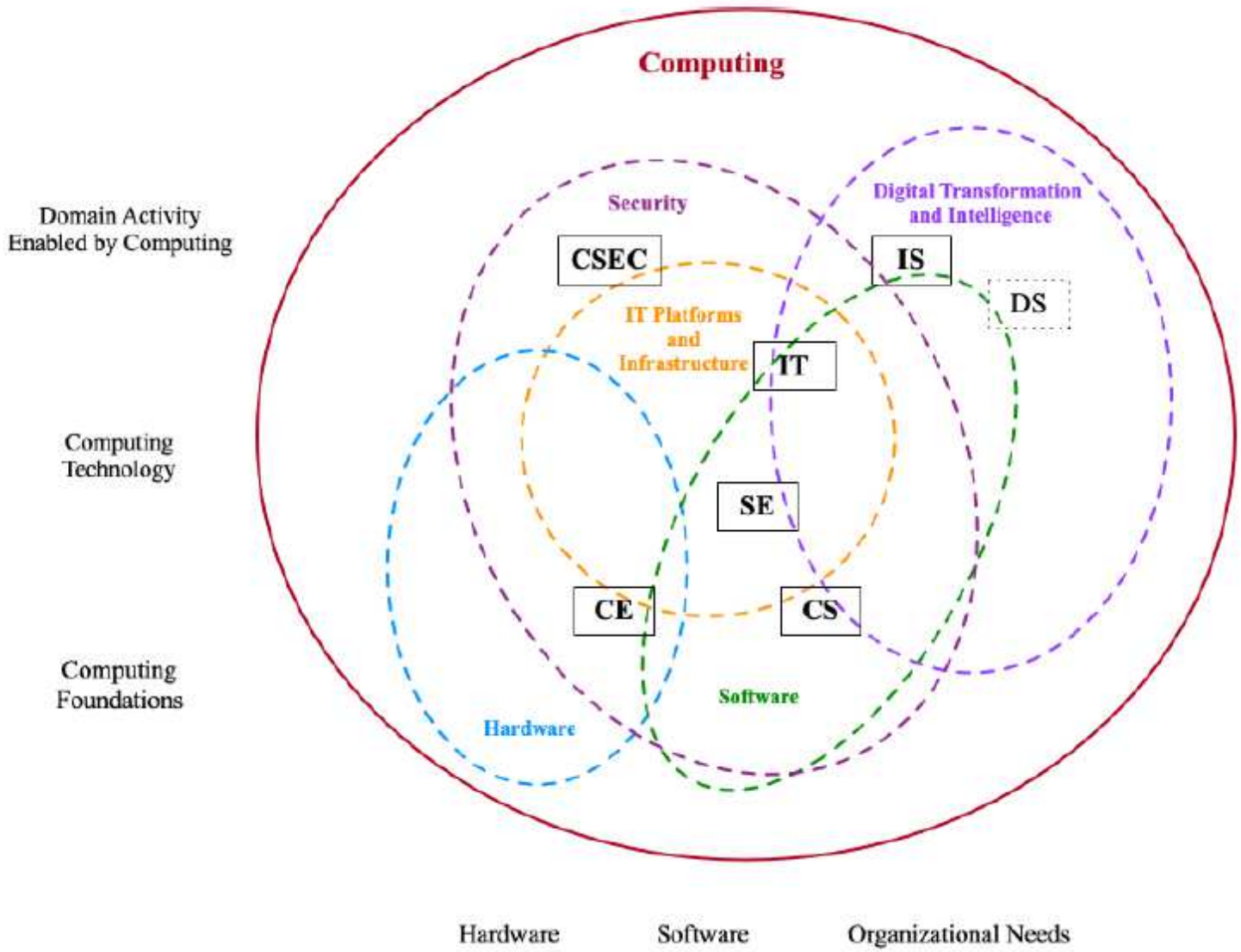


Figure 2.2. A contemporary view of the landscape of computing education
Legend: Curricular reports: CE=computer engineering; CS=computer science; CSEC=cybersecurity; IS=information systems; IT=information technology; SE=software engineering; DS=data science (under development).

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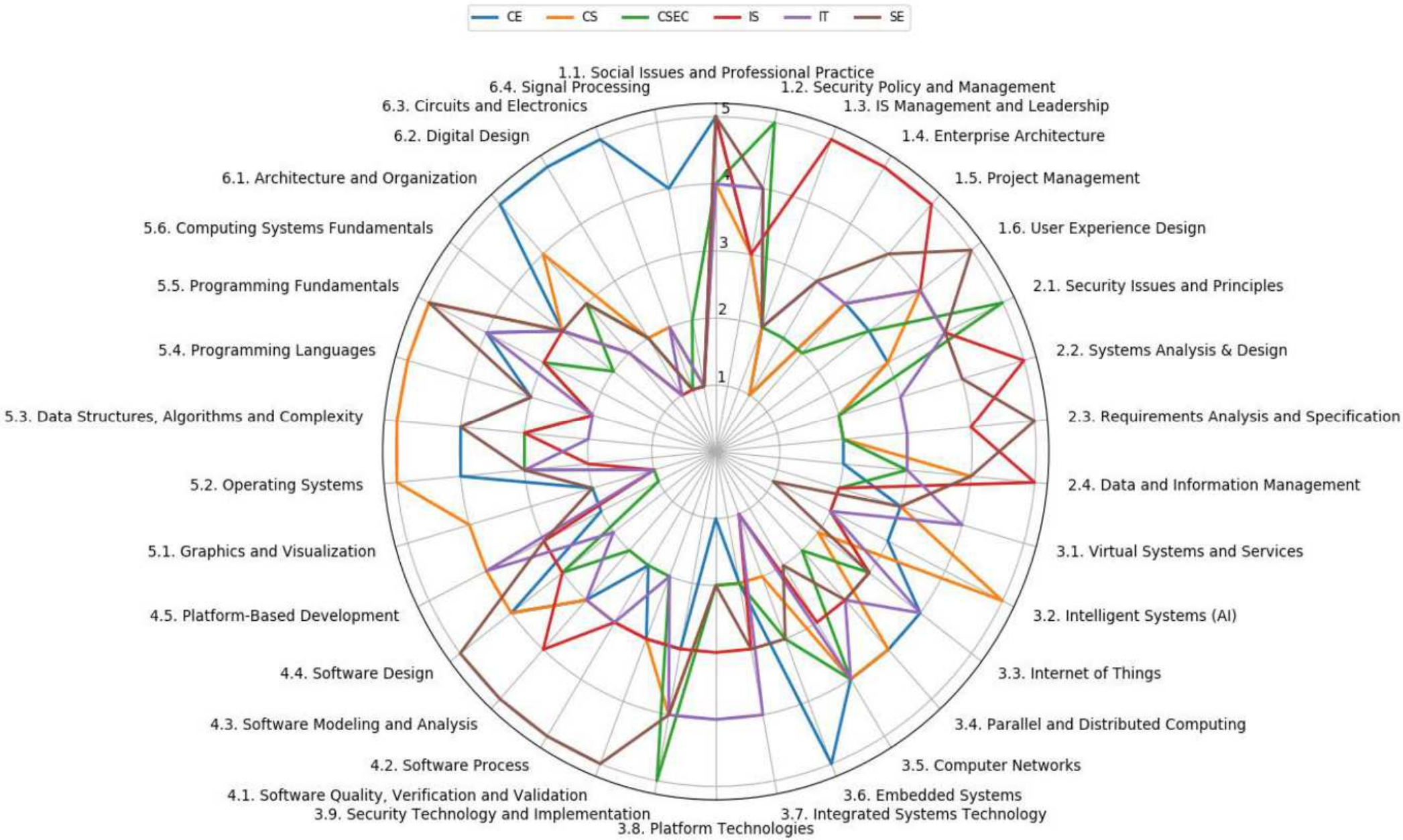
Table 5.3 Landscape of Computing Knowledge

		CE		CS		CSEC		IS		IT		SE	
		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
1. Users and Organizations	1.1. Social Issues and Professional Practice	2	5	2	4	2	4	3	5	2	4	3	5
	1.2. Security Policy and Management	1	3	2	3	4	5	2	3	2	4	2	4
	1.3. IS Management and Leadership	0	2	0	2	1	2	4	5	1	2	1	2
	1.4. Enterprise Architecture	0	1	0	1	1	2	3	5	1	3	1	3
	1.5. Project Management	1	3	2	3	1	2	4	5	2	3	2	4
	1.6. User Experience Design	1	3	2	4	1	3	2	4	3	4	3	5
2. Systems Modeling	2.1. Security Issues and Principles	2	3	2	3	4	5	2	4	3	4	2	4
	2.2. Systems Analysis & Design	1	2	1	2	1	2	4	5	1	3	2	4
	2.3. Requirements Analysis and Specification	1	2	1	2	0	2	2	4	1	3	3	5
	2.4. Data and Information Management	1	2	2	4	2	3	3	5	2	3	2	4
3. Systems Architecture and Infrastructure	3.1. Virtual Systems and Services	1	3	1	3	1	2	1	2	3	4	1	3
	3.2. Intelligent Systems (AI)	1	3	3	5	1	2	1	2	1	2	0	1
	3.3. Internet of Things	2	4	0	2	1	3	1	3	2	4	1	3
	3.4. Parallel and Distributed Computing	2	4	2	4	1	2	1	3	1	3	2	3
	3.5. Computer Networks	2	4	2	4	2	4	1	3	3	4	2	2
	3.6. Embedded Systems	3	5	0	2	1	3	0	1	0	1	0	3
	3.7. Integrated Systems Technology	1	2	0	2	0	2	1	3	3	4	1	3
	3.8. Platform Technologies	0	1	1	2	1	2	1	3	2	4	0	2
	3.9. Security Technology and Implementation	2	3	2	4	4	5	1	3	2	4	2	4
4. Software Development	4.1. Software Quality, Verification and Validation	1	3	1	3	1	2	1	3	1	2	3	5
	4.2. Software Process	1	2	1	3	0	2	1	3	1	3	3	5
	4.3. Software Modeling and Analysis	1	3	1	3	1	2	2	4	1	3	4	5
	4.4. Software Design	2	4	2	4	1	3	1	3	1	2	4	5
	4.5. Platform-Based Development	0	2	2	4	0	1	1	3	2	4	1	3
5. Software Fundamentals	5.1. Graphics and Visualization	1	2	2	4	0	1	1	1	0	1	0	2
	5.2. Operating Systems	2	4	3	5	2	3	1	2	1	3	1	3
	5.3. Data Structures, Algorithms and Complexity	2	4	4	5	1	3	1	3	1	2	2	4
	5.4. Programming Languages	2	3	3	5	1	2	1	2	1	2	2	3
	5.5. Programming Fundamentals	2	4	4	5	2	3	1	3	2	4	3	5
	5.6. Computing Systems Fundamentals	2	3	2	3	1	2	2	3	1	3	2	3
6. Hardware	6.1. Architecture and Organization	4	5	3	4	1	3	1	2	1	2	1	3
	6.2. Digital Design	4	5	1	2	0	2	0	1	0	1	0	2
	6.3. Circuits and Electronics	4	5	1	2	0	1	0	1	1	2	0	1
	6.4. Signal Processing	3	4	0	1	0	2	0	1	0	1	0	1

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